

# Solutions to Dummit & Foote's Abstract Algebra 3<sup>rd</sup> Edition

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0.3  $\mathbb{Z}/n\mathbb{Z}$ : The Integers Modulo  $n$ **Exercise 0.3.1**

Write down explicitly all the elements in the residue classes of  $\mathbb{Z}/18\mathbb{Z}$ .

**Solution.** Note that  $\bar{x} = \{x + 18k \mid k \in \mathbb{Z}\}$ . Explicitly, we have:

$$\begin{array}{lll} \bar{0} = \{0, 18, -18, 36, -36, \dots\} & \bar{1} = \{1, 19, -17, 37, -35, \dots\} & \bar{2} = \{2, 20, -16, 38, -34, \dots\} \\ \bar{3} = \{3, 21, -15, 39, -33, \dots\} & \bar{4} = \{4, 22, -14, 40, -32, \dots\} & \bar{5} = \{5, 23, -13, 41, -31, \dots\} \\ \bar{6} = \{6, 24, -12, 42, -30, \dots\} & \bar{7} = \{7, 25, -11, 43, -29, \dots\} & \bar{8} = \{8, 26, -10, 44, -28, \dots\} \\ \bar{9} = \{9, 27, -9, 45, -27, \dots\} & \bar{10} = \{10, 28, -8, 46, -26, \dots\} & \bar{11} = \{11, 29, -7, 47, -25, \dots\} \\ \bar{12} = \{12, 30, -6, 48, -24, \dots\} & \bar{13} = \{13, 31, -5, 49, -23, \dots\} & \bar{14} = \{14, 32, -4, 50, -22, \dots\} \\ \bar{15} = \{15, 33, -3, 51, -21, \dots\} & \bar{16} = \{16, 34, -2, 52, -20, \dots\} & \bar{17} = \{17, 35, -1, 53, -19, \dots\} \quad \blacksquare \end{array}$$

**Exercise 0.3.2**

Prove that the distinct equivalence classes in  $\mathbb{Z}/n\mathbb{Z}$  are precisely  $\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}$ .

**D&F HINT.** Use the Division Algorithm.

**Solution.** Let  $m \in \mathbb{Z}$ . By the Division Algorithm, there exists unique  $q, r \in \mathbb{Z}$  such that

$$m = nq + r, \quad 0 \leq r < n.$$

Then  $m \equiv r \pmod{n}$ , so  $m \in \bar{r}$ . By definition,  $\bar{r}$  is one of  $\bar{0}, \bar{1}, \dots, \overline{n-1}$ . Thus, every equivalence class in  $\mathbb{Z}/n\mathbb{Z}$  is one of  $\bar{0}, \bar{1}, \dots, \overline{n-1}$ . Moreover, they are all distinct. Otherwise, if  $\bar{x} = \bar{y}$  for  $0 \leq x, y < n$ , then  $x \equiv y \pmod{n}$ , or  $n \mid (x - y)$ . However, since  $-n < x - y < n$ , we must have  $x - y = 0$ , or  $x = y$ .  $\blacksquare$

**Exercise 0.3.3**

Prove that if

$$a = a_n 10^n + a_{n-1} 10^{n-1} + \dots + a_1 10 + a_0$$

is any positive integer, then

$$a \equiv (a_n + a_{n-1} + \dots + a_1 + a_0) \pmod{9}.$$

**D&F HINT.** Use the fact that  $10 \equiv 1 \pmod{9}$ .

**D&F NOTE.** Note that this is the usual arithmetic rule that the remainder after division by 9 is the same as the sum of the decimal digits mod 9—in particular an integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

**Solution.** Using the hint, then

$$\begin{aligned} a &\equiv (a_n 10^n + a_{n-1} 10^{n-1} + \dots + a_1 10 + a_0) \pmod{9} \\ &\equiv (a_n 1^n + a_{n-1} 1^{n-1} + \dots + a_1 1 + a_0) \pmod{9} \\ &= (a_n + a_{n-1} + \dots + a_1 + a_0) \pmod{9} \quad \blacksquare \end{aligned}$$

**Exercise 0.3.4**

Compute the remainder when  $37^{100}$  is divided by 29.

**Solution.** Consider the following calculations mod 29, where we note that  $37 \equiv 8 \pmod{29}$ :

$$8^2 \pmod{29} \equiv 6$$

$$8^8 \pmod{29} \equiv 20$$

$$8^{32} \pmod{29} \equiv 7$$

$$8^4 \pmod{29} \equiv 7$$

$$8^{16} \pmod{29} \equiv 23$$

$$8^{64} \pmod{29} \equiv 20$$

Then  $8^{100} \equiv 8^{64} \cdot 8^{32} \cdot 8^4 \equiv 20 \cdot 7 \cdot 7 \equiv 23 \pmod{29}$ . ■

### Exercise 0.3.5

Compute the last two digits of  $9^{1500}$ .

**D&F NOTE.** One may also do this by considering the binomial expansion of  $(10 - 1)^{1500}$ , but we will maintain the approach of using modular arithmetic.

**Solution.** Doing repeated squaring, we have

$$9^2 \pmod{100} \equiv 81$$

$$9^8 \pmod{100} \equiv 21$$

$$9^{32} \pmod{100} \equiv 21$$

$$9^4 \pmod{100} \equiv 61$$

$$9^{16} \pmod{100} \equiv 61$$

$$9^{64} \pmod{100} \equiv 41$$

Observe that  $9^{64} \equiv 9^4 \equiv 61 \pmod{100}$ . Then  $9^{60} \equiv 1 \pmod{100}$ , so that  $9^{1500} \equiv 9^{25 \cdot 60} \equiv 1^{25} \equiv 1 \pmod{100}$ . Thus, the last two digits of  $9^{1500}$  are 01. ■

### Exercise 0.3.6

Prove that the square of the elements in  $\mathbb{Z}/4\mathbb{Z}$  are just  $\bar{0}$  and  $\bar{1}$ .

**Solution.**  $0^2 \equiv 0 \pmod{4}$ ,  $1^2 \equiv 1 \pmod{4}$ ,  $2^2 \equiv 0 \pmod{4}$ , and  $3^2 \equiv 1 \pmod{4}$ . ■

### Exercise 0.3.7

Prove that for any integers  $a$  and  $b$  that  $a^2 + b^2$  never leaves a remainder of 3 when divided by 4.

**D&F HINT.** Use Exercise 0.3.6.

**Solution.** By Exercise 0.3.6,  $a^2$  and  $b^2$  are either 0 or 1 mod 4, so their sum cannot be 3 mod 4. ■

### Exercise 0.3.8

Prove that the equation  $a^2 + b^2 = 3c^2$  has no solutions in nonzero integers  $a$ ,  $b$  and  $c$ .

**D&F HINT.** Consider the equation mod 4 as in the previous two exercises and show that  $a$ ,  $b$  and  $c$  would all have to be divisible by 2. Then each of  $a^2$ ,  $b^2$  and  $c^2$  has a factor of 4 and by dividing through by 4 show that there would be a smaller set of solutions to the original equation. Iterate to reach a contradiction.

**Solution.** Considering the equation modulo 4, we know that  $a^2 + b^2 \equiv 0, 1, 2 \pmod{4}$  by Exercise 0.3.7, but  $3c^2 \equiv 0, 3 \pmod{4}$  by Exercise 0.3.6. Hence, both sides must be congruent to 0 mod 4. Since  $3c^2 \equiv 0 \pmod{4}$ , we have  $c^2 \equiv 0 \pmod{4}$ , so that  $c$  is even. Then  $3c^2$  has a factor of 4, so that  $a^2 + b^2$  has a factor of 4, and hence both  $a$  and  $b$  are even. Divide each of  $a$ ,  $b$  and  $c$  by 2 to get a smaller solution to the original equation. By infinite descent, we can iterate the process to obtain no solution in nonzero integers  $a$ ,  $b$  and  $c$ . ■

**Exercise 0.3.9**

Prove that the square of any odd integer always leaves a remainder of 1 when divided by 8.

**Solution.** Put  $a = 2k + 1$  for  $k \in \mathbb{Z}$ . Then  $a^2 = 4k(k + 1) + 1 \equiv 1 \pmod{8}$  since  $k(k + 1)$  is always even. ■

**Exercise 0.3.10**

Prove that the number of elements of  $(\mathbb{Z}/n\mathbb{Z})^\times$  is  $\varphi(n)$ , where  $\varphi$  denotes the Euler  $\varphi$ -function.

**Solution.** It suffices to show that  $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$  if and only if  $(a, n) = 1$ .

( $\Rightarrow$ ) If  $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$ , then there exists  $\bar{b} \in \mathbb{Z}/n\mathbb{Z}$  such that  $\bar{a} \cdot \bar{b} = \bar{1}$ . Then  $ab \equiv 1 \pmod{n}$  implies the existence of  $k \in \mathbb{Z}$  such that  $ab - 1 = kn$ , or  $ab - kn = 1$ . Thus,  $(a, n) = 1$ .

( $\Leftarrow$ ) By  $(a, n) = 1$ , there exists  $x, y \in \mathbb{Z}$  such that  $ax + ny = 1$ , or  $ax = 1 - ny$ . We then have  $ax \equiv 1 \pmod{n}$ , so that  $\bar{a} \cdot \bar{x} = \bar{1}$ . Thus,  $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$ . ■

**Exercise 0.3.11**

Prove that if  $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$ , then  $\bar{a} \cdot \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$ .

**Solution.** Since  $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$ , there exist  $\bar{c}, \bar{d} \in \mathbb{Z}/n\mathbb{Z}$  such that  $\bar{a} \cdot \bar{c} = \bar{b} \cdot \bar{d} = \bar{1}$ . Then

$$(\bar{a} \cdot \bar{b})(\bar{c} \cdot \bar{d}) = (\bar{a} \cdot \bar{c})(\bar{b} \cdot \bar{d}) = \bar{1} \cdot \bar{1} = \bar{1},$$

so that  $\bar{a} \cdot \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$ . ■

**Exercise 0.3.12**

Let  $n \in \mathbb{Z}, n > 1$  and let  $a \in \mathbb{Z}$  with  $1 \leq a \leq n$ . Prove if  $a$  and  $n$  are not relatively prime, there exists an integer  $b$  with  $1 \leq b < n$  such that  $ab \equiv 0 \pmod{n}$  and deduce that there cannot be an integer  $c$  such that  $ac \equiv 1 \pmod{n}$ .

**Solution.** Let  $(a, n) = d > 1$ . Then  $a/d, n/d \in \mathbb{Z}$ . Put  $b = n/d$ , so that  $1 \leq b < n$  and

$$ab = a(n/d) = (a/d)n \equiv 0 \pmod{n}.$$

If there existed some  $c \in \mathbb{Z}$  such that  $ac \equiv 1 \pmod{n}$ , then  $(ac)b \equiv b \pmod{n}$ . However,  $ab \equiv 0 \pmod{n}$ , so that  $0 \equiv b \pmod{n}$ , which is impossible since  $1 \leq b < n$ . ■

**Exercise 0.3.13**

Let  $n \in \mathbb{Z}, n > 1$  and let  $a \in \mathbb{Z}$  with  $1 \leq a \leq n$ . Prove that if  $a$  and  $n$  are relatively prime then there is an integer  $c$  such that  $ac \equiv 1 \pmod{n}$ .

**D&F HINT.** Use the fact that the GCD of two integers is a  $\mathbb{Z}$ -linear combination of the integers.

**Solution.** Since  $(a, n) = 1$ , there exists  $c, x \in \mathbb{Z}$  such that  $ac + nx = 1$ . Then  $ac \equiv 1 \pmod{n}$ . ■

**Exercise 0.3.14**

Conclude from the previous two exercises that  $(\mathbb{Z}/n\mathbb{Z})^\times$  is the set of elements  $\bar{a}$  of  $\mathbb{Z}/n\mathbb{Z}$  with  $(a, n) = 1$  and hence prove Proposition 4. Verify this directly in the case  $n = 12$ .

**Solution.** The previous two exercises show that  $a$  and  $n$  are relatively prime if and only if there exists  $b$  such that  $ab \equiv 1 \pmod{n}$ , which is exactly the proposition. For  $n = 12$ , the elements 1, 5, 7, 11 are relatively prime to 12, so that  $(\mathbb{Z}/12\mathbb{Z})^\times = \{\bar{1}, \bar{5}, \bar{7}, \bar{11}\}$  whose inverses are  $\bar{1}, \bar{5}, \bar{7}, \bar{11}$  respectively. ■

#### Exercise 0.3.15

For each of the following pairs of integers  $a$  and  $n$ , show that  $a$  is relatively prime to  $n$  and determine the multiplicative inverse of  $\bar{a}$  in  $\mathbb{Z}/n\mathbb{Z}$ .

- (a)  $a = 13, n = 20$
- (b)  $a = 69, n = 89$
- (c)  $a = 1891, n = 3797$
- (d)  $a = 6003722857, n = 77695236973$

**Solution.** Use the Euclidean Algorithm to find  $x, y \in \mathbb{Z}$  such that  $ax + ny = 1$  for each pair  $(a, n)$ . Then  $ax \equiv 1 \pmod{n}$ , and  $x$  is the multiplicative inverse of  $\bar{a}$  in  $\mathbb{Z}/n\mathbb{Z}$ .

- (a) 17.
- (b) 40.
- (c) 253.
- (d) 77695236753. ■

#### Exercise 0.3.16

Write a computer program to add and multiply mod  $n$ , for any  $n$  given as input. The output of these operations should be the least residues of the sums and products of two integers. Also include the feature that if  $(a, n) = 1$ , an integer  $c$  between 1 and  $n - 1$  such that  $\bar{a} \cdot \bar{c} = \bar{1}$  may be printed on request.

**D&F NOTE.** Your program should not, of course, simply quote “mod” functions already built into many systems.

**Solution.** Please see `modular_arithmetic.py` for the solution. ■